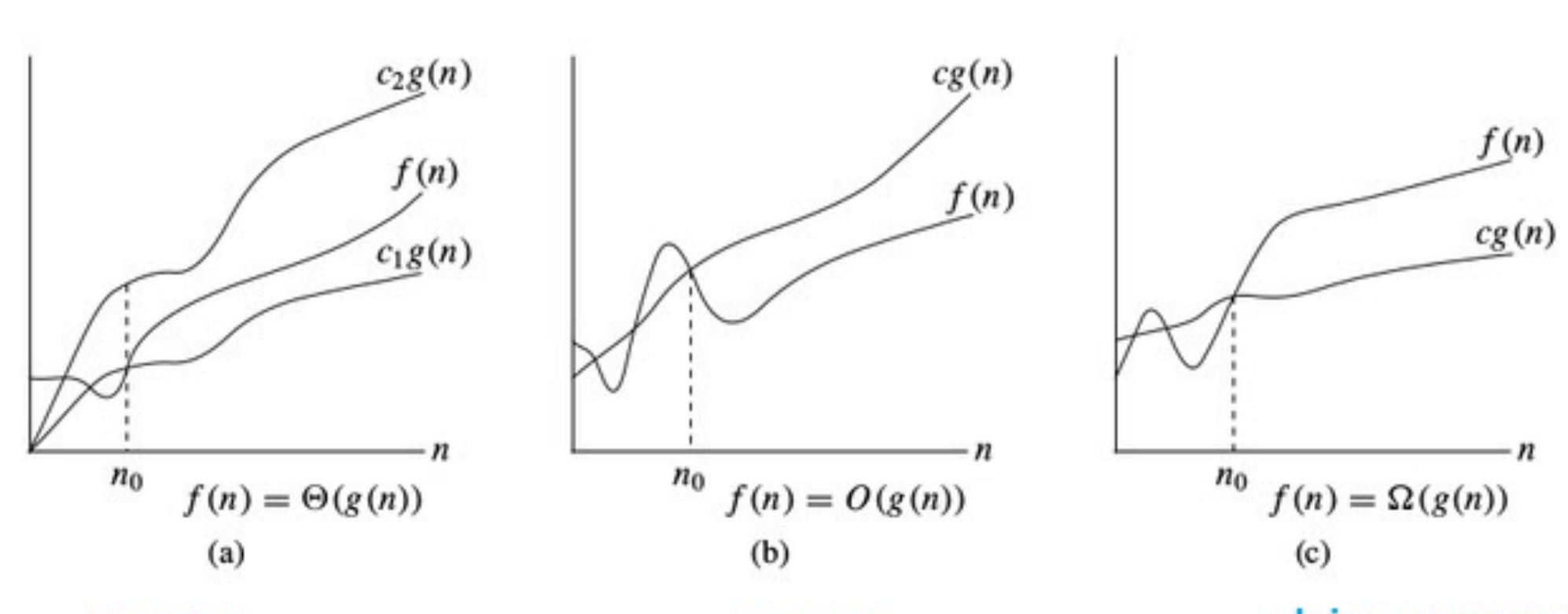
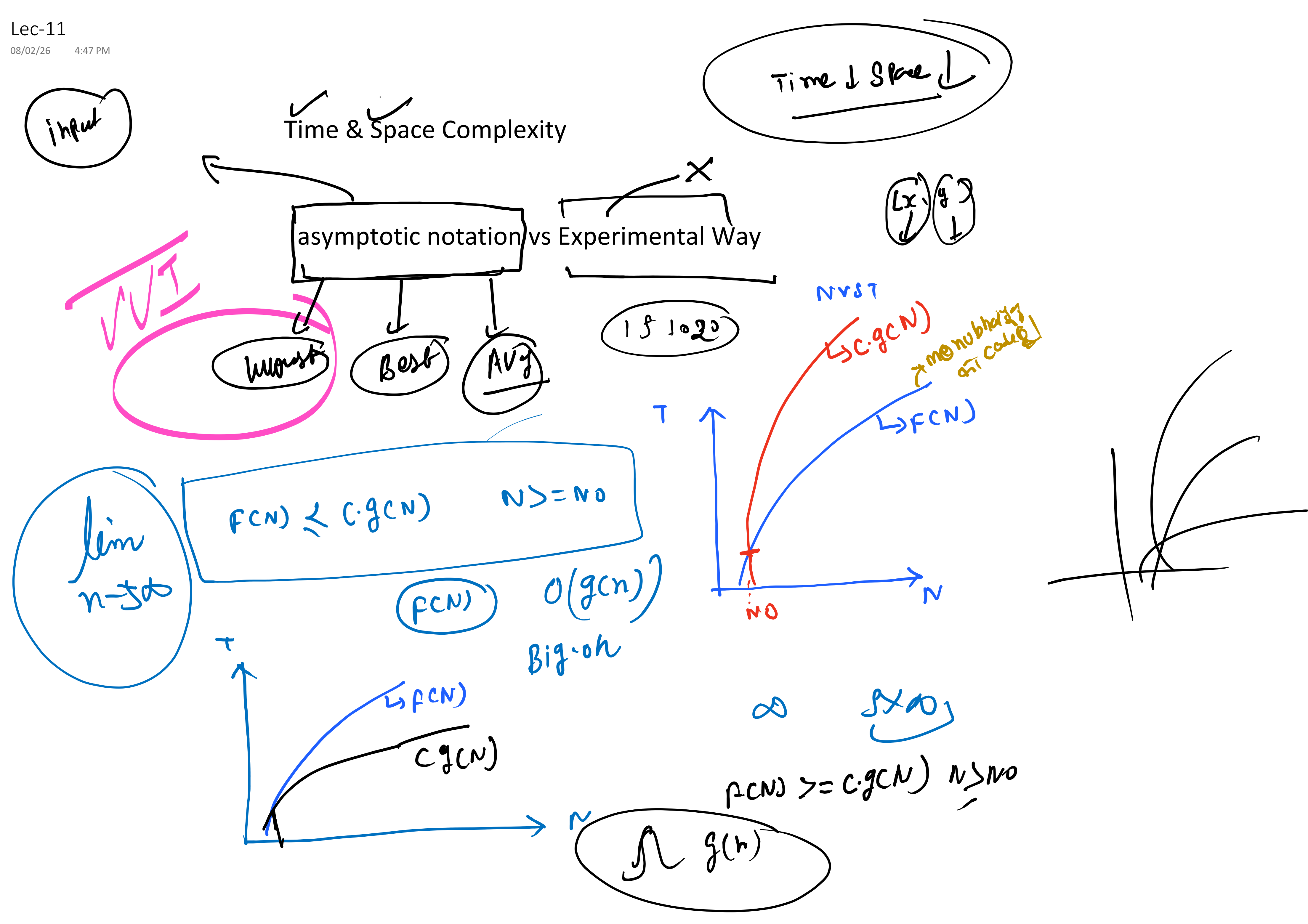




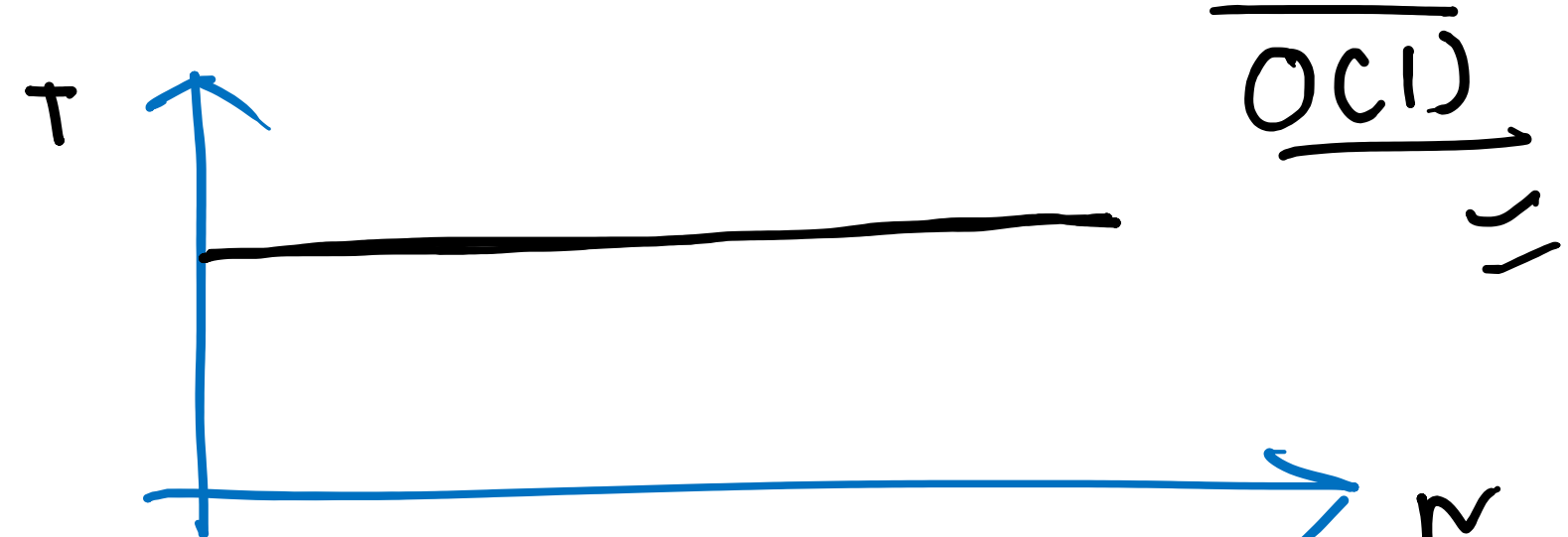
$$f(N) \leq C \cdot g(N)$$

$$f(n) = 2n^2 + 4n + 7n^3 + 8$$

$$P(n) = 2^n + 7(2^n \log n) + 7n \log n$$

$C = 22$
 $\delta(n) = n^5$

$$2n^2 + 4n + 7n^3 \leq 22n^5$$



```
public static int Search(int[] arr, int item) {
    for (int i = 0; i < arr.length; i++) {
        if (arr[i] == item) {
            return i;
        }
    }
    return -1;
}
```

i=0 i=1 i=2 i=3
1 1 1 1

while (n) $\rightarrow$ while $(i \leq n)$
 $i = i + n$

```
while (i <= n) {
    System.out.println("Hey");
    i *= 2;
}
```

$i = 1 \quad i = 2 \quad i = 2^2 \quad i = 2^3 \quad \dots \quad i = 2^{k-1}$
 $1 \quad \quad \quad 1 \quad \quad \quad 1 \quad \quad \quad 1$

 $(k) \rightarrow \log_2 N$

```
while (i <= n) {
    System.out.println("Hey");
    i += 2;
    i += 3;
}
```

20

$i=0 \quad i=5 \quad i=10 \quad i=15 \quad \dots \quad i=100$

1 1 1 1 1

3;

$\frac{N}{S}$

$$\underline{O(n)}$$

```
for (i = 1; i <= n; i++) {  
    for (int j = 1; j <= n; j++) {  
        System.out.println("hey");  
    }  
}
```

$i=1 \quad i=2 \quad i=3 \quad i=4 \quad i=5 \quad \dots \quad i=n$
 $\sim \quad \sim \quad \sim \quad \sim \quad \sim \quad \sim$

$T(n) = n \times n = n^2$

A hand-drawn diagram illustrating a for loop. A speech bubble on the left contains the word "loop". A bracket above the loop body is labeled with a circled "5". An arrow points from the "i++" increment part of the loop header to a small box containing the number "6".

```
for (i = 1; i * i <= n; i++) {
    System.out.println("hey");
}
```

$$l^2 \leq N$$

$$l \leq \sqrt{N}$$

```
for (i = 1; i <= n; i++) {
    for (int j = 1; j <= i * i; j++) {
        for (k = 1; k <= n / 2; k++) {
            System.out.println("hey");
        }
    }
}
```

$i=1 \quad i=2 \quad i=3 \quad i=4 \quad i=5 \quad \dots \quad i=n$

$\frac{2}{1} \quad \frac{2}{2} \times \frac{2}{2} \quad \frac{2^2 \times 2}{2} \quad \frac{4^2 \times 2}{2} \quad \frac{5^2 \times 2}{2} \quad \dots \quad \frac{n^2 \times 2}{2}$

$$\sum_{n=1}^{\infty} \left[\frac{n(n+1)(2n+1)}{6} \right]$$

```
for (i = n / 2; i <= n; i++) {  
    for (int j = 1; j <= n / 2; j++) {  
        for (k = 1; k <= n; k = k * 2) {  
            System.out.println("hey");  
        }  
    }  
}
```

$$\frac{n_1 \times \log(n)}{2} \times \frac{n}{2} = \frac{n^2 \log(n)}{2}$$

```
for (i = 1; i <= n; i++) {  
    for (int j = 1; j <= n; j += i) {  
        System.out.println("hey");  
    }  
}
```

$i=1$ $i=2$ $i=3$ $i=4$ $i=5$ $\dots i=n$
 N $\frac{N}{2}$ $\frac{N}{3}$ $\frac{N}{4}$ $\frac{N}{5}$ $\frac{N}{n}$ $\uparrow \log(n)$
 $N \left[1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{n} \right]$

$$\int \log(N) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{N}$$
$$K_{\text{NW}} = 10^5 \times 10^5 = 10^{10} > 10^8$$

Red, Pink, and

Rev

0 and 2

7.

```
zero = 0
while (i = TWO) {
```

$\text{swap}(arr[i], arr[j])$
 $\text{swap}(arr[i], arr[j])$
 $\text{swap}(arr[i], arr[j])$

3
abc

2